DROUGHT · HEAVY RAIN · HEAT · NORTHERN KENYA

Ukame Watch

Early warning for eight arid counties, built so a county officer can act before a season fails

The last eight rainy seasons, eight-county average rainfall as % of normal



43% Oct–Dec 22 | **133%** Mar–May 23 | **174%** Oct–Dec 23 | **136%** Mar–May 24 | **70%** Oct–Dec 24 | **111%** Mar–May 25 | **42%** Oct–Dec 25 | **129%** Mar–May 26

Zetech Alumni × Student Hackathon 2026 · ML & Data Science · Climate Sentinel

Patrick Wangai · Faith Githaiga · Monchari Leah · Dedan Mugambi

REAL-WORLD PROBLEM AND SECTOR RELEVANCE

Drought, then flood, in the same counties

5.6 million

people in Turkana, Marsabit, Wajir, Mandera, Garissa, Isiolo, Samburu and Kitui

KNBS 2019 census

5 droughts

documented since 2000; three were declared national disasters

FAO, World Weather Attribution, ACAPS

135,125

people displaced by the late-2023 floods; Garissa, Mandera and Wajir worst hit

UN OCHA, 20 November 2023

Before a season fails, a county disaster officer needs four answers: **Is this season unusual? What comes next? Who is exposed? What do we prepare?**

SOLUTION RELEVANCE, INNOVATION AND SUITABILITY

One board answers the four questions

Is this season unusual?

Rainfall as % of the 1991–2020 normal, and an NDMA-style drought level for each county

What comes next?

A 3-month outlook tested against two baselines, and a live 7-day heavy-rain and heat watch

Who is exposed?

Census population per county, and the trend in hot days since 1991

What do we prepare?

A cited action playbook, and an offline AI that drafts an advisory a named officer must approve

WHAT IS NEW

A forecast that reports where it fails, and an AI that cannot invent a number, skip the playbook or bypass a person

LIVE TODAY · 13 NOVEMBER 2026

Samburu is at drought Alarm; two counties face heavy rain

26% of normal rain

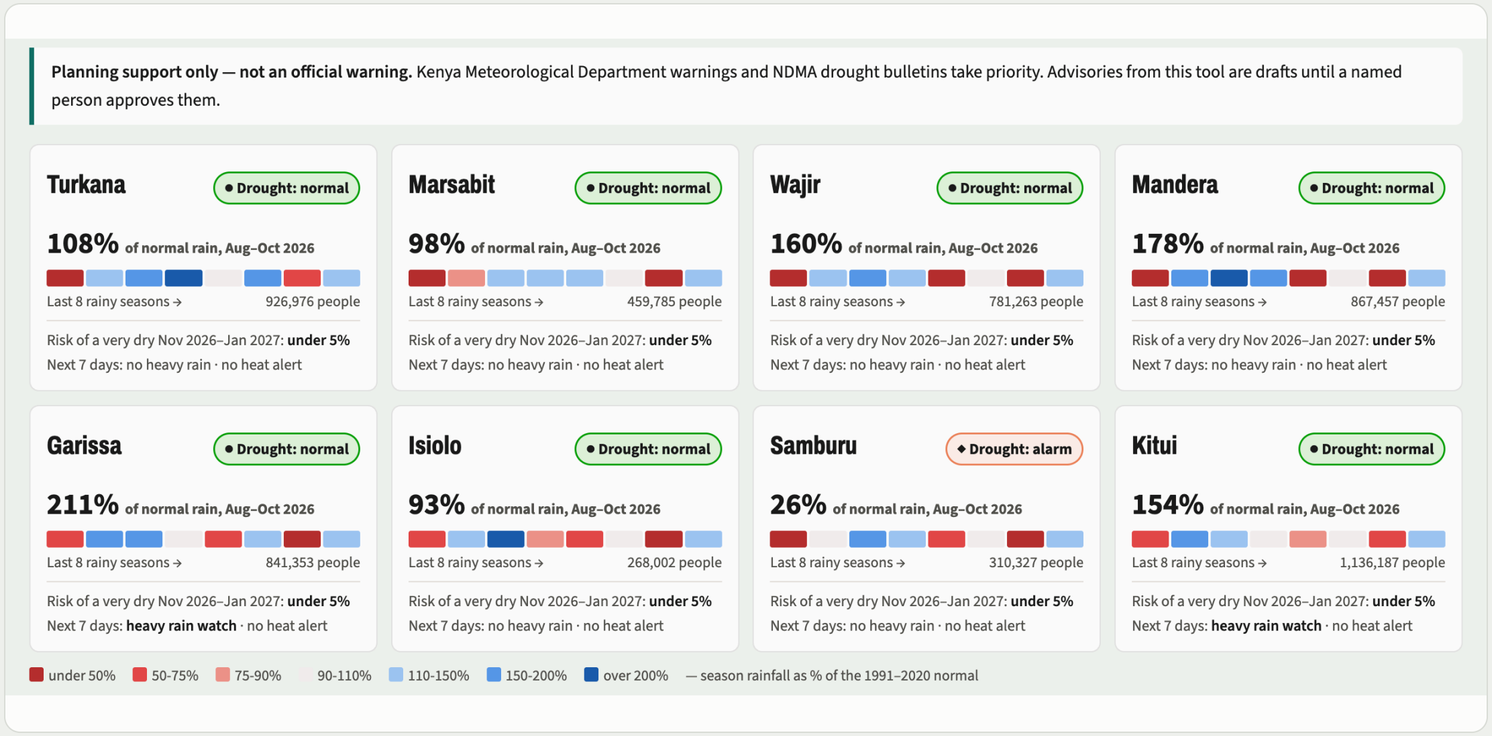
Samburu, Aug–Oct 2026: drought level Alarm

Heavy-rain watch

Next 7 days: Garissa and Kitui

Under 5%

risk of a very dry Nov–Jan, with a strong El Niño (+2.46) and positive Indian Ocean Dipole (+0.95)



Live demo: patrickwangai.github.io/ukame-watch · weather data through 9 Nov 2026

MEASURABLE OUTCOMES AND EVALUATION

It catches the disasters Kenya already lived through

10 of 10 documented droughts and floods, 1997–2024, flagged by our indicators

6 of 6 worst droughts and heavy-rain periods in 35 years of data match a documented disaster

+0.12 forecast skill for Oct–Dec rains on unseen years 2017–2026 (Brier skill score vs climatology); persistence scores −0.01

+5 to +10 extra hot days per decade, statistically significant in 7 of 8 counties

Where it fails: no skill for the Mar–May long rains (−0.05). The app says so on screen.

3 of the 10 events were added after the indicator flagged them, then verified independently; they are labelled.

TECHNICAL ARCHITECTURE AND TECHNOLOGY STACK

From free public data to an approved advisory

Free data

NASA POWER daily since 1991, ERA5, NOAA El Niño and Indian Ocean indices, census

→

Quality control

11 impossible rainfall days caught by cross-checking a second dataset

→

Indicators and forecast

Drought, heavy rain and heat; one logistic model per season

→

Local AI with guardrails

Qwen 2.5 3B, six tools, every number checked in code

→

Person approves

Drafts stay drafts until a named officer approves; full audit log

93 automated tests

One command rebuilds every number in about a minute

Runs offline on an 8 GB laptop

No GPU, no paid services

Python · pandas · scikit-learn · pymannkendall · GeoPandas · Streamlit · Plotly · Ollama · stlite on GitHub Pages

We designed for what can go wrong

Risk	What we built
Rainfall data is model-based	Cross-checked with ERA5; 11 impossible days found and fixed
The AI invents or overstates	Every number and level word checked; “no risk” blocked; a person approves
Forecasts look too certain	Below 5% shown as “under 5%”; Mar–May labelled no skill
Mistaken for official warnings	On every screen: KMD and NDMA warnings take priority
Wrong population figures	Census used; WorldPop is 3 times too high in Mandera
Running costs or servers fail	Free public data, free hosting, works offline on a laptop

Full list: docs/limitations.md in the repository

REAL-WORLD IMPLEMENTATION, FEASIBILITY AND SCALABILITY

Free to run today, ready to scale

Works today

- Zero running cost: free public data, free hosting
- Runs offline on an 8 GB laptop; nothing leaves the device
- One command rebuilds every number from scratch
- Open source, with 93 automated tests

Next 12 months

1. Pilot with one County Steering Group in the 2027 long rains
2. Add Kenya Meteorological Department station rainfall
3. Extend to every county NDMA monitors: settings plus validation
4. Swahili summaries by SMS and radio for pastoralists

SDG ALIGNMENT AND SOCIETAL IMPACT

Earlier action against Kenya’s costliest climate hazards

SDG 13.1

Climate resilience: county signals before a season fails

SDG 2.4

Food security: time for water repairs and timely livestock offtake

SDG 1.5

Poverty: protect household assets in some of Kenya’s poorest counties

SDG 11.5

Disasters: heavy-rain watches prompt early warning and evacuation plans

HOW A PILOT WILL MEASURE IMPACT

Lead time: days from signal to county action

Trust: share of drafts approved without edits

Use: officer-rated usefulness each month

Also serves livestock and agriculture officers, and humanitarian partners such as the Kenya Red Cross

Earlier warnings for 5.6 million people, and honest about their limits

Try it now: patrickwangai.github.io/ukame-watch

Code, data sources and limitations: github.com/PatrickWangai/ukame-watch